

Title : Statistical & Machine Learning Analysis of Airbnb Listings in NYC (2024): Pricing, Demand, and Market Structure

Project Description and Research Method: This project will analyze Airbnb listings in New York City (2024) using statistical and machine learning methods to study pricing, demand, and host strategies. Descriptive and inferential statistics will be applied to identify significant differences across boroughs and room types, while supervised learning models will predict prices and classify demand levels. Unsupervised learning techniques such as clustering and PCA will be used to segment listings and uncover market patterns. The project emphasizes statistical rigor, model interpretability, and data-driven insights relevant for both academic research and practical decision-making.

Problem Statements

Descriptive + Inferential

- Q1: How do prices differ across boroughs and room types?
→ Hypothesis testing (ANOVA / Kruskal-Wallis), effect size estimates, confidence intervals.
- Q2: Does host strategy (single vs. multi-listing) significantly affect average price or review activity?
→ Two-sample t-tests or Mann-Whitney, regression with interaction terms.

Supervised Learning

- Q3: Can we predict listing **price** from features (room type, location, bedrooms, reviews)?
→ Linear regression, Ridge/Lasso, Random Forest, Gradient Boosted Trees. Compare statistical and ML models.
- Q4: Can we classify listings into **high-demand vs. low-demand** based on review activity?
→ Logistic regression, Decision Trees, SVM. ROC/AUC, calibration plots.

Unsupervised Learning

- Q5: What clusters of listings exist (based on price, room type, host behavior, reviews, ratings)?
→ KMeans, Gaussian Mixture Models, Hierarchical clustering.
→ Validate clusters using silhouette score, gap statistic.
- Q6: Can neighborhoods be grouped into statistically similar market segments?
→ PCA + clustering on aggregated neighborhood features.

Project Description:

This project aims to conduct a comprehensive statistical and machine learning analysis of Airbnb listings in New York City using a dataset of over 20,000 listings and 22 attributes. The study will explore pricing dynamics, host strategies, and guest demand patterns across boroughs and room types. The primary objectives are to (1) describe and visualize market characteristics using descriptive statistics, (2) test hypotheses on price and demand differences through inferential statistical methods, (3) build supervised learning models to predict listing price and classify demand levels, and (4) apply unsupervised learning techniques to segment hosts, listings, and neighborhoods. This project will integrate statistical rigor with machine learning interpretability to generate insights that are academically sound and practically relevant for urban short-term rental markets.

Research Methods:

The methodology combines statistical data science techniques with supervised and unsupervised learning. Descriptive statistics and visualizations will summarize key features of the dataset. Inferential statistics—including hypothesis testing (ANOVA, chi-square, t-tests), regression modeling, and confidence intervals—will be used to assess differences and relationships among boroughs, room types, and host strategies. Supervised learning approaches such as linear regression, penalized regression, logistic regression, decision trees, and ensemble methods will be employed to predict price and demand proxies, with model performance evaluated using RMSE, MAE, accuracy, and ROC-AUC. Unsupervised learning methods, including clustering (KMeans, hierarchical, Gaussian mixtures) and dimensionality reduction (PCA), will identify latent market segments and listing patterns. Results will be interpreted with a focus on statistical validity, model explainability, and implications for both hosts and policymakers.